NOT MEASUREMENT SENSITIVE

MIL-C-29567
30 March 1989

MILITARY SPECIFICATION

# CLOTH, LAMINATED, STRETCHABLE, WATERPROOF AND MOISTURE VAPOR PERMEABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1 SCOPE

1 1 Scope This specification covers the requirements for a spandex knit face cloth laminated to a layer of expanded polytetrafluoroethylene film covered by a backing cloth of spandex knit

## 2 APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-P-1136 | Packaging of Coated (Plastic, Rubber) and Laminated Fabrics |

MILITARY

| | |
|---|---|
| MIL-T-5624 | Turbine Fuel, Aviation, Grades JP-4 and JP-5 |
| MIL-A 8243 | Anti Icing and Deicing Defrosting Fluids |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-191 | Textile Test Methods |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 8305

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STANDARDS (Cont'd)

MILITARY

MIL-STD-831 Test Reports, Preparation of

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099 )

2 1.2 Other Government publications The following other Government publication forms a part of this document to the extent specified herein Unless specified, the issues are those cited in the solicitation

PUBLICATIONS

FEDERAL REGULATIONS

Rules and Regulations under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580 )

2 2 Non-Government publications The following documents form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6 2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 96 Water Vapor Transmission of Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103 )

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC 116 Colorfastness to Crocking, Rotary Vertical Crockmeter Method

AATCC 135 Dimensional Changes in Automatic Home Launderings of Woven and Knit Fabrics

(Nongoverment standards and other publications are normally available from the organizations which prepare or distribute the documents These documents also may be available in or through libraries or other informational services )

2 3 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained



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3 REQUIREMENTS

3 1 First Article When specified (see 6.2),a sample of the laminated cloth shall be subject to first article inspection (see 4 4 and 6.3).

3 2 Materials.

3 2 1 Base Cloth

3 2 1 1 Face The face cloth shall be a 64 gauge 80% nylon 6,6/20% spandex blend weighing 5 7 ± 0 6 ounces per square yard Minimum wales count shall be 40 and minimum course count shall be 49. The color shall be Sage Green No. 1590.

3 2 1 2 Middle layer. The middle layer shall include a microporous expanded polytetrafluoroethylene film weighing 0.5 ± 0.2 ounces per square yard and an adhesive suitable for the purpose intended

3 2 1 3 Backing The backing cloth shall be an 80% nylon 6,6/20% spandex blend weighing 1 8 ± 0 3 ounces per square yard Minimum wales count shall be 41 and minimum course shall be 57 The color shall be Sage Green No 1590

3 2 2 Laminated cloth The base cloths shall be laminated together in such a manner that the assembled (finished) cloth shall comply with the requirements specified in Table I, when tested as specified in 4 6 (see Figure 1)

TABLE I Laminated cloth and component properties

| Characteristics | Requirements |
|---|---|
| Weight, ounces per square yard | 9 +1.5 -0 0 |
| Colorfastness to Perspiration | Equal to or better than, the standard sample or equal to, or better than good |
| Laundering (after 3 cycles) | Equal to, or better than, the standard sample or equal to, or better than good |
| Wet and Dry Crocking | Equal to, or better than, the standard sample or equal to, or better than 3-4, on the AATCC Chromatic Transference Scale |
| Delamination after agitation | no delamination 1/ |
| Dimensional Stability (after 5 cycles) | 8% wale 6% course |



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TABLE I Laminated cloth and component properties (cont'd)

| Characteristics | Requirements |
|---|---|
| Film Overall Weight | see 4 6 1 |
| Moisture vapor transmission rate(MVTR) $gm/m^2/24$ hrs , | minimum |
| Water method (upright) | 2/ 750 |
| Inverted water method | 3/ 4,000 |
| Water Permeability, Maximum | |
| Initial | no failures |
| After synthetic perspiration | no failures |
| After exposure to aircraft fluids | no failures |
| Percent Elongation at 12 lbs , (min) | 60 |
| Percent Recovery at 12 lbs , (min) | 70 |
| Resistance to organic liquid | |
| Initial | no wetting |
| After 3 launderings | no wetting |
| Spray Rating | |
| Initial (min) | 4/ 100, 90, 90 |
| After 3 launderings (min) | 4/ 90, 90, 80 |

1/ The cloth shall be considered delaminated if the separation exceeds 0 25 inches in two perpendicular directions

2/ This minimum value shall be obtained by averaging all of the determinations obtained from the individual specimens taken from all of the sample units as a group No single specimen determination shall be less than 650

3/ This minimum value shall be obtained by averaging all of the determinations obtained from the individual specimens taken from all of the sample units as a group No single specimen determination shall be less than 3,500

4/ The results of the three individual determinations on the sample unit for spray rating shall be equal to or better than the minimum requirement given

3 2 3 Water repellency The laminated cloth shall be given an approved Quarpel water repellent treatment on the face side of the cloth and shall conform to the water repellency requirements The use of materials other than approved water repellents are prohibited (see 6 4)

3 3 Dimension of the laminated cloth



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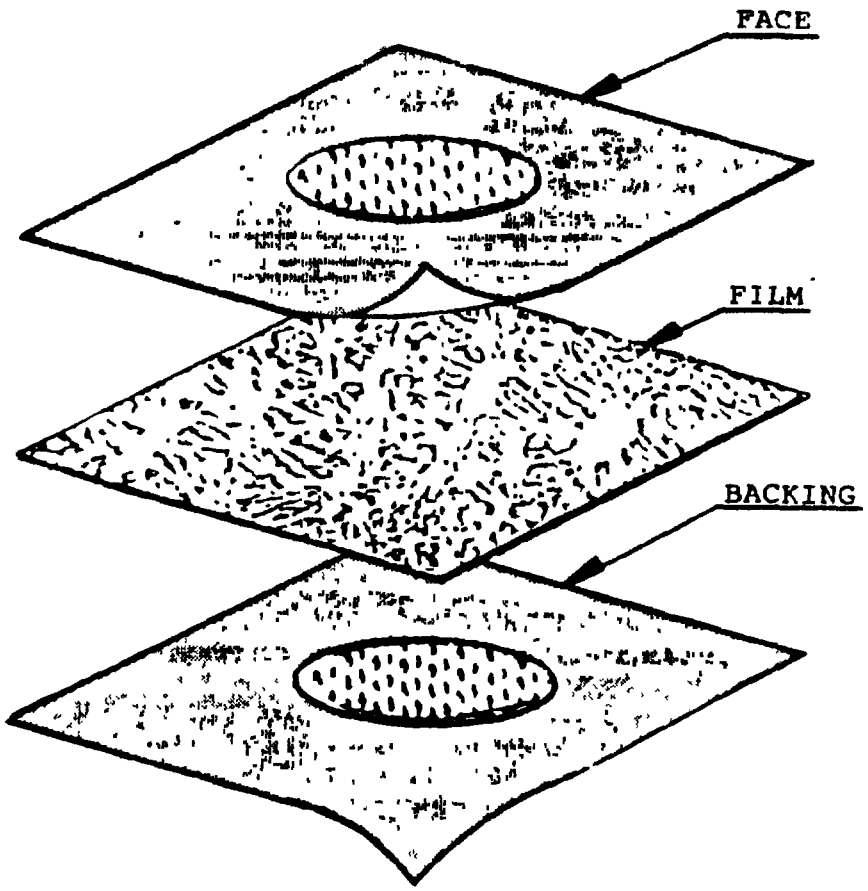



FIGURE 1 Exploded view of laminated cloth



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3 3 1 Length and put-up The laminated cloth shall be put up on rolls as specified by the procuring activity (see 6 2 1). If not specified by the procuring activity, the minimum length of any one roll shall be 50 linear yards, the maximum number of pieces per roll shall be three and no single piece shall be less than 10 yards

3 3 2 Width Unless otherwise specified by the procuring activity (see 6 2 1), the width of the laminated cloth shall be not less than 54 inches

3 4 Fiber identification Each roll shall be labeled and ticketed for fiber content in accordance with the Textile Fiber Products Identification Act

3 5 Workmanship The cloth shall be uniform in quality and shall be free from irregularities or any defects that could adversely affect their performance, reliability, or durability Because of the use of this cloth for the manufacture of emergency and life support items, the importance of providing a product of uniformly excellent quality cannot be overemphasized The cloth shall conform to the quality and grade of product established by this specification

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements

4 1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of the defective material

4 1.2 Test equipment and inspection facilities The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections

4 2 Classification of inspections The inspection requirements specified herein are classified as follows



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a First article inspection The first article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4 4)

b Quality conformance inspection Quality conformance inspection consists of examinations and tests performed on the individual products or lots to determine comformance of the products or lots with the requirements set forth in this specification (see 4 5)

4.3 Inspection conditions Unless otherwise specified herein, all testing shall be performed under standard atmospheric conditions and performed on specimens in moisture equilibrium under standard atmospheric conditions specified in FED-STD-191 or ASTM as applicable

4 4 First Article

4 4 1 First article inspection. First article inspection of the laminated cloth shall consist of examinations and tests for all of the requirements of this specification

4 4 2 First article sample Unless otherwise specified, as soon as practical after the award of the contract or order, the manufacturer shall submit seven linear yards of laminated cloth The sample shall be representative of the construction, workmanship, components and materials to be used during production When a contractor is in continuous production of this laminated cloth from contract to contract, submission of further first article inspection samples on the new contract, may be waived at the discretion of acquiring activity (see 6 2 1) Approval of the first article inspection samples or the waiving of the first article inspection does not waive the requirements for performing the quality conformance inspection The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1). A test report, containing complete test data for each component required herein, shall be prepared in accordance with MIL-STD-831 The test report shall accompany the first article sample

4.4.3 First article report. Upon completion of the first article inspection program, the Government activity responsible for conducting the inspection program shall report to the contracting officer the results of the inspection program, with appropriate recommendations.

4.4 4 Disposition of first article samples Upon completion of the first article inspection, recommendations and comments pertinent for use in monitoring production will be forwarded by the Government activity responsible for the inspection program (see 6 2 1) to the contracting officer The samples will be consumed or destroyed during testing and shall not be considered as part of the quantity to be delivered under the contract or order

4 5 Quality conformance inspection The quality conformance inspection shall consist of the examinations in 4 5 1 and 4 5 2 and all the tests required for the laminated cloth for each lot



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4 5 1 Examination of the end item The end item shall be examined in accordance with the classification of defects and acceptable quality levels (AQLs) set forth in the contract (see 6.2 and 6 4).

4.5.1.1 Yard-by-yard examination. Every yard of every roll in the lot shall be inspected and visual defects classified as listed in Table II. The cloth shall be inspected on the face side, except every tenth roll shall be examined on the back side. The defects found shall be counted regardless of their proximity to each other except where two or more of the same defects represent a single local condition occurring within one square yard of the cloth, in which case only one defect shall be counted A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The entire yardage of the lot shall be visually examined.

TABLE II Classification of defects for the laminated cloth.

| Defects | Classification | |
|---|---|---|
| | Major | Minor |
| Blisters or lumps· | | |
| - 1/4 square inch or less in combined directions, clearly visible 1/ | | X |
| - Over 1/4 square inch in combined directions, clearly visible 1/ | X | |
| Laminating faults | | |
| - Creases | X | |
| - Folds - winding on tubes | | X |
| - Any evidence of delamniation | X | |
| Any cut, tear or hole | X | |
| Floats, slubs or skips - multiple 1/2 inch or more in either warp or filling direction | | X |
| Foreign matter incorporated | X | |
| Stains on surface: | | |
| - Any spot or stain (compound, oil dirt) over 1 inch in length in any direction, clearly visible 1/ | | X |
| Streaks: | | |
| - Shade streaks on face side, clearly visible 1/ | X | |
| Any tackiness | X | |

1/ The term "clearly visible" shall be interpreted as meaning clearly visible at normal examination distance (approximately 3 feet) under an overhead illumination.



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4 5 1 2 Overall examination Each roll in the lot shall be examined for overall defects The lot shall be considered unacceptable if any of the defects listed in Table III are present

TABLE III Overall defects

| |
|---|
| Selvage curled, folded, rolled, or slack continuously or intermittently throughout the piece |
| Overall uncleanness or objectionable odor |
| Uneven lamination throughout the piece. |
| Off-shade, shaded end to end side to side, side to center or throughout the piece |
| Traceable numbering system, which can be traced to the date of lamination, missing from end of roll or identification tag. |

4 5 1 3 Examination for length and width.

4.5 1 3 1 Individual roll for length and width Every roll in the lot shall be examined for the defects listed below The lot shall be considered unacceptable if any of these defects are present

a Gross length of roll less than minimum specified (see 3 3 1)

b Gross length of roll less than gross length marked on ticket.

c Length of shortest piece on roll less than minimum specified (see 3 3 1)

d Width of roll less than minimum specified (see 3 3 2)

e Any roll containing more than the maximum number of pieces specified (see 3 3.1)

4.5 1.3.2 Total yardage in lot. The lot shall be unacceptable if the total of the actual gross length of pieces in the roll are less than the total of the gross lengths marked on the tickets

4 5 1 4 Examination for compliance with Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this Act shall be defective The lot shall be unacceptable if two or more of these defects occur

4 5 2 Inspection of packaging The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of PPP-P-1136



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4.6 Testing of the end item The methods of testing specified in 4 6 1, 4 6 2 and as listed in Table IV shall be followed Unless otherwise specified, test methods listed in Table IV are contained in FED-STD-191 The physical and chemical values specified in Table I except where otherwise specified, shall apply to the average of the determinations made on the individual specimens from the sample units, as specified in the applicable test methods All individual values shall be shown and utilized in expressing the final results

4 6 1 Film overall weight test The polytetrafluoroethylene film shall be tested in accordance with Method 5041 of FED-STD-191 except that each specimen shall be 25 square inches and shall be cut in diagonal fashion from each sample unit and may be tested under ambient conditions The specimens shall be equally spaced across the full width of the sample unit no closer than 2 inches to the edges of the sample unit The distance between the top of a specimen and the bottom of the next specimen shall be 3 inches The sample unit shall be 1 1/2 yards full width of the plastic film The lot shall be rejected if any specimen weighs less than 0 3 or greater than 0 7 ounces per square yard, or if the difference in weight between any two specimens in the sample unit is greater than 0 25 ounces per square yard The sample size shall be stated in the contract (see 6 2 and 6 5)

4 6 2 Physical and chemical testing The lot size shall be expressed in units of one yard Specimens for each test shall be cut in a diagonal fashion across the entire width of the sample unit so that no two specimens are taken from areas that overlap in the width or lengthwise direction The sample unit for test purposes shall be seven continuous yards, full width, of the laminated cloth The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified (see Table I)

4 6 2 1 Dimensional stability Specimens shall be 1 yard by full width Load size shall be 4 pounds Dummy load shall consist of comparable material. Bench marks shall be 18 inches. Do not press

4.6 2.2 Colorfastness

4 6 2 2 1 Colorfastness to perspiration The colorfastness to perspiration shall be done in accordance with FED-STD-191 method 5680 using only the acid solution. Plexiglass plates 4 1/2 inches by 2 1/2 inches by 1/8 inch shall be acceptable. The size of the color transfer cloth shall be 2 1/4 inches by 2 1/4 inches Evaluate only that stain evident on the Nylon 6,6 stripe of the multifiber test cloth

4 6.2.2.2 Colorfastness to laundering. Specimen weight shall be 4+0 1 gram The specimen assemblies shall be pressed after the third cycle using a flat bed press at 275-302°F for 15 seconds

4.6.2.2.3 Colorfastness to wet and dry crocking The color and appearance of the laminated cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500± 200K, with illumination of 100 ± 20 foot candles and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K



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TABLE IV. Test Methods.

| Characteristics | Test Paragraph | FED-STD-191 Test Method | Other Test Method |
|---|---|---|---|
| Weight | | 5041 | |
| Dimensional Stability | 4.6.2 1 | | AATCC 135 I, III,B |
| Colorfastness to perspiration | 4 6.2 2 1 | 5680 | |
| Colorfastness to laundering | 4.6.2.2.2 | 5614 | |
| Colorfastness to wet and dry crocking | 4.6.2.2.3 | | AATCC 116 |
| MVTR Upright | 4 6.2.3 1 | | ASTM E 96 |
| MVTR Inverted | 4.6 2 3.2 | | ASTM E 96 |
| Water Permeability<br>Initial<br>After synthetic perspiration<br>After aircraft fluids | 4 6 2 4<br>4 6 2 4 1<br>4.6.2.4 2<br>4 6 2 4 3 | 5516 | |
| Percent Elongation | 4.6 2 5 | Section 4 | |
| Percent Recovery | 4.6.2.5 | Section 4 | |
| Resistance to organic liquid<br>Initial<br>After 3 launderings | 4 6 2 6<br>4 6 2 6.1<br>4 6 2.6.2 | | <br><br>AATCC 135 I, III B |
| Spray Rating<br>Initial<br>After 3 launderings | 4 6 2 7<br>4.6.2.7<br>4 6.2 7 | 5526<br>5526<br>5526 | <br>AATCC 135 I, III B<br>AATCC 135 I, III B |
| Delamination after agitation | 4 6 2 8 | | |

4.6.2.3 Moisture Vapor Transmission Rate (MVTR).

4 6.2 3.1 Moisture Vapor Transmission Upright. The moisture vapor transmission shall be tested in accordance with ASTM E 96 The back side of the laminated cloth shall face the water The free stream air velocity shall be 550 ±50 fpm as measured in the center of the airflow cress section at the downstream end of the chamber's test section The test shall be for 24 hours and weight measurements shall be taken only at the start and completion of the



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test At the start of the 24 hour test period, the air gap between the water surface and the back of the specimen shall be 3/4 inch ±1/16 inch Five specimens shall be tested Test chamber temperature shall be 73 4°F ±1°F and relative humidity shall be 50 ±2 percent

4 6.2.3.2 Moisture Vapor Transmission Inverted The moisture vapor transmission shall be tested in accordance with ASTM E 96. The back side of the laminated cloth shall face the water The free stream air velocity shall be 550 ±50 fpm as measured in the center of the airflow cross section at the downstream end of the chamber's test section The test shall run for 2 hours and weight measurements shall be taken only at the start and completion of the test. Five specimens shall be tested Specimens shall be sealed in any manner which prevents wicks and/or leakage of water out of the cup. Test chamber shall be 73 4°F ±1°F and relative humidity shall be 50 ±2 percent

4 6.2 4 Water permeability

4.6 2.4 1 Initial The initial water permeability shall be measured as specified in method 5516 of FED-STD-191, except that the face side of the laminated cloth shall contact the water The hydrostatic head shall be 50 centimeters and shall be held for 3 minutes. The report shall only include the appearance of water Leakage is defined as the appearance of water any place within the 4 1/2 inch diameter test area The test may be performed using any device which tests the same specimen area at equivalent pressure. In cases of dispute, the apparatus decribed in Method 5516 of FED-STD-191 shall be used The specimen size shall accomodate the equipment being used

4.6 2 4 2 After synthetic perspiration One specimen per sample unit shall be tested for water permeability after exposure to synthetic perspiration The specimens shall not be less than six inches in diameter The test cups (see 6 7) shall accomodate this size specimen and shall have a depth of at least one inch The cups shall be sealed against leakage Synthetic pespiration shall be prepared by stirring 3 grams (gm) sodium chloride, 1 gm predigested protein (see Table V), 1 gm n-propyl proprinate, and 0 5 gm lecithin (phosphatidyl choline) into 500 milliliters (ml) distilled water. The solution shall be stirred continuously and heated to 50°C, then covered and cooled to 35°C The solution shall be stirred in such a way that any solid particles are suspended in solution and poured to a level 0.5 inch in the test cup The cup shall be inverted to allow the synthetic perspiration to evaporate through the sample according to ASTM E 96, Method BW. The airflow control may be waived After the solution has evaporated through the sample such that no more than 0 125 inch of solution remains, the specimen shall be removed from the cup, rinsed in warm water, dried and tested for leakage as specified in 4.6 2 4 1



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TABLE V Predigested protein contents.

| Ingredient | Milligrams (mg) |
|---|---|
| Lysine | 82.5 |
| Histidine | 27.5 |
| Arginine | 40 0 |
| Aspartic Acid | 72.5 |
| Threonine | 42.5 |
| Serine | 50 0 |
| Glutamic acid | 197 5 |
| Proline | 92 5 |
| Glycine | 22 5 |
| Alanine | 28 7 |
| Cystine | 4 7 |
| Valine | 66 2 |
| Methionine | 30 0 |
| Isolencine | 53.8 |
| Leucine | 87.5 |
| Tyrosine | 51.3 |
| Phenylalanine | 48 8 |
| Tryptophane | 18.8 |

4 6.2 4 3 After aircraft fluids One specimen per sample shall be tested after exposure to aircraft fluids. The specimens shall be 6 inches minimum in diameter. Specimens shall be placed on a flat surface and 1 ml of JP-4 (jet fuel conforming to MIL-T-5624) shall be spread over the middle, followed by 1 ml of deicing fluid (conforming to MIL-A-8243) Place the horizontal specimens flat in an air circulating oven such that air circulates on both sides of the specimen at 50°C for 30 minutes. Remove from oven and test immediately for water permeability



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4 6 2 5 Percent elongation and recovery The tests for percent elongation and percent recovery shall be as follows·

Specimen size shall be 3 inches by 8 inches; 8 inches being the test direction Three warp specimens shall be cut straight across (not on a diagonal) Fold specimens in half to create a 3 inch by 4 inch rectangle and sew the 3 inch edges together using a lock stitch at 15 stitches per inch and a 1 inch seam allowance; back stitch each edge Condition specimens according to section 4 of FED-STD-191. A constant rate-of extension tester consisting of a straining mechanism, load and elongation recording mechanism, and C-clamps shall be used See Figures 2 and 3 for C-clamp dimensions as recommended by the Lycra Products Group, E.I. Du Pont Company, Textile Research Lab, Chestnut Run, Wilmington, Delaware. C-clamp edges which contact the specimen should be rounded or should be padded with one layer of at least 5 9 ounce stretch Lycra knit Guage length is 3 inches. Set crosshead speed at 5 inches per minute. Loop specimen through openings in C-clamps Cycle specimen between zero and 12 pounds three times without stopping Measure the extension from the zero pound point to the twelve pound point of pull number three Measure the recovery along the zero point from the start of the first pull to the start of the third pull

4 6 2 6 Resistance to organic liquid.

4.6 2 6 1 Initial The initial test for resistance to organic liquid shall be as follows

Place a small specimen of the cloth on a smooth horizontal surface, face up Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After 30 seconds examine the specimen under light at an angle. Absence of light reflectance at the cloth drop interface shall be taken as evidence of wetting Three specimens (or areas) taken at various locations across the sample unit shall be tested Evidence of wetting on one or more specimens shall be considered a test failure

4.6.2.7 Spray rating after three launderings The specimens shall be laundered in accordance with method AATCC 135 I, III, B Load size shall be four pounds Dummy load shall consist of comparable material. Do not press

4 6 2.8 Delamination after agitation One specimen, 14 inches by full width, shall be selected from each sample unit and tested for delamination after 24 hours of continuous agitation The specimens shall be agitated using the "normal" cycle in an automatic home laundering machine except that the washing machine shall be capable of continuous agitation. The water level shall be maintained at 18 ± 0.5 gallons, and water temperature shall be 32 ± 9°C. The load shall be 2 ± 0.2 pounds Upon completion of 24 hours of continuous agitation, and while still wet, the specimen shall be evaluated for delamination as described in footnote 1 of Table I

4 6.3 Retesting



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4 6 3 1 Retesting for Direct Shipment to the Government Individual rolls, from lots which have been rejected for any of the physical requirements listed in Table I may be retested once for the failed characteristic Each roll in the rejected lot shall be inspected individually A sample unit of the cloth, of sufficient length to perform the test for the failed characteristic, shall be taken from the end of that roll The sample unit shall be tested for the failed characteristics and if the sample unit passes the test requirements for the characteristic in question, the roll will be considered acceptable

4 6 3 2 Retesting for Shipment to a Contractor Unless otherwise specified, if defective yardage can be identified without testing, the defective rolls shall be removed from the lot and the lot resubmitted for testing for the failed characteristic(s) using the number of samples specified in 4 5

5 PACKAGING

5 1 Preservation Preservation shall be level A or C as specified (see 6 2)

5 1 1 Levels A and C The cloth shall be preserved in accordance with the applicable requirements of PPP-P-1136

5 2 Packing Packing shall be level A, B or C as specified (see 6 2)

5 2 1 Levels A, B and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5 3 Marking In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with PPP-P-1130 The following precautionary marking shall appear on all containers

"STORE IN A COOL DRY PLACE "

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory )

6.1 Intended use. The laminated cloth is intended for use in the fabrication of waterproof socks for continuous wear anti-exposure clothing

6.2 Ordering data.

6.2 1 Acquisition requirements Acquisition documents should specify the following

a. Title, number and date of this specification

b Quantity



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c Length and put up requirements (see 3 3 1)

d Width requirements (see 3.3 2)

e Whether first article is waived (see 4 4 1)

f Name and address of the first article inspection laboratory (see 4 4 1), including the Government Activity responsible for conducting the inspection program (see 4 4 2)

g Selection of applicable levels of preservation and packing (see 5 1)

h Acceptable Quality level (see 6 4)

i Sample Size (see 6.5)

6 3 First article When a first article inspection is required, the item should be a first production item as required by the 4 4 1 The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection for those bidders offering a product which has been previously acquired or tested by the Govenment and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the contract.

6 4 Examination of the end item AQL If not stated in the contract the AQL for this examination (4.5 1 1) shall be 15 5 defects per 100 yards

6.5 Sample size If not stated in the contract, the sample size for the film overall weight test (4 6 1) and the physical and chemical testing (4 6.2), should be as follows

| Lot Size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 to 22,000 | 3 |
| 22,001 and over | 5 |

6 6 Quarpel water repellent The "Quarpel type" water repellent treatment consist of the co-application of an emulsified fluorocarbon and a fluorocarbon extender Approval of components and combinations is the responsibility of U.S Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 and is based on extensive tests, including those for toxicity which are not set forth in this document. Because of the time necessary to conduct full evaluation (approximately 6 month), only those chemical treatments already approved and so listed in the invitation for bids or request for proposals shall be considered acceptable for the related procurement

6 7 Test cups Contamination test chambers, 6 inches in diameter, manufactured by Pesce Lab Sales, P O Box 235, Kenneth Square, PA 19348 have been successfully used for the leakage after perspiration. Equivalent test cups may be used



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6 8 Standard sample For access to standard sample, address the contracting activity issuing the invitation for bids

6 9 Subject term (key word) listing

Cloth
First Article
Laminated
Lycra knit
Moisture vapor permeable
Polytetraflouroethylene film
Stretch sock
Waterproof

Custodians
Navy - AS
Air Force - 99

Review
Air Force - 11, 82
DLA - CT

Preparing Activity
Navy - AS
(DOD Project No 8305-0273)



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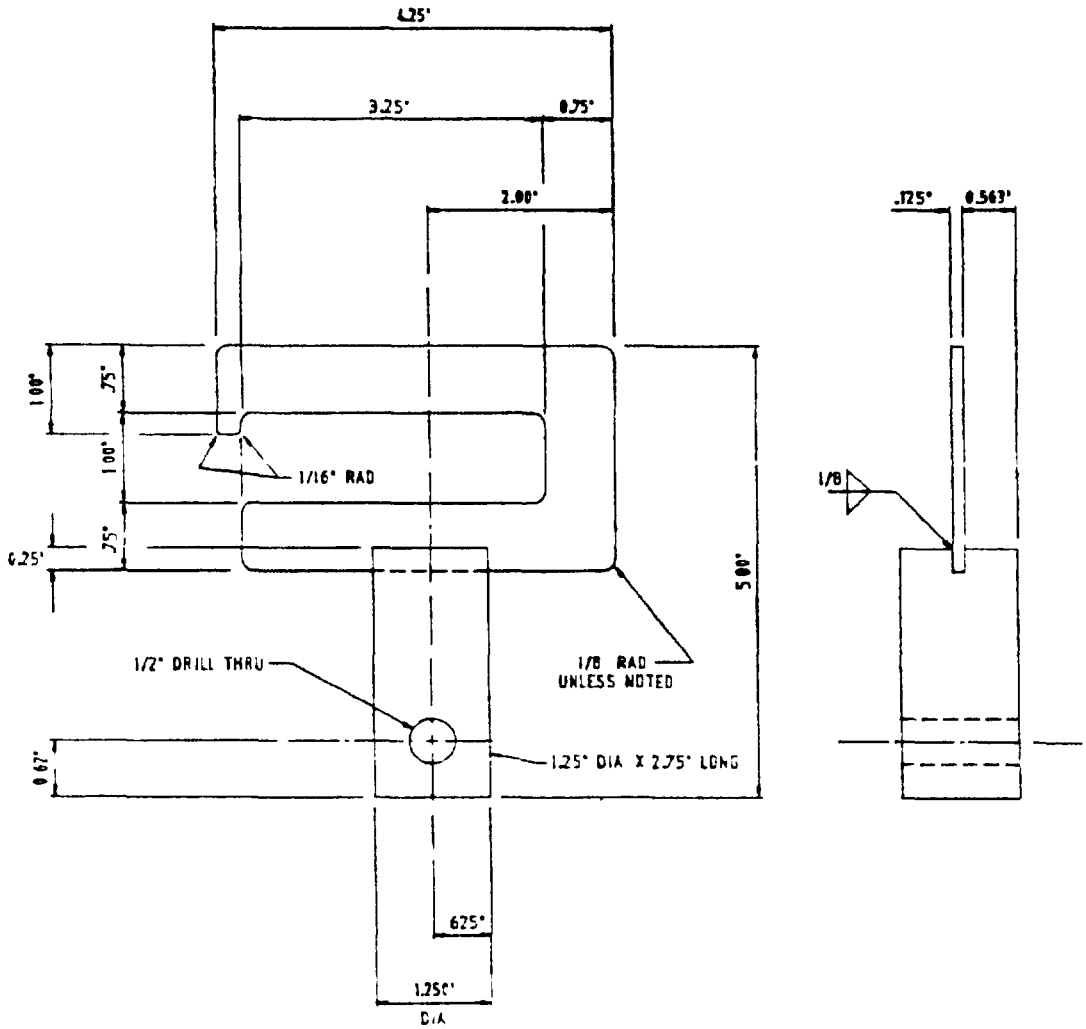



FIGURE 2 C-Clamp Dimensions - Large Shaft



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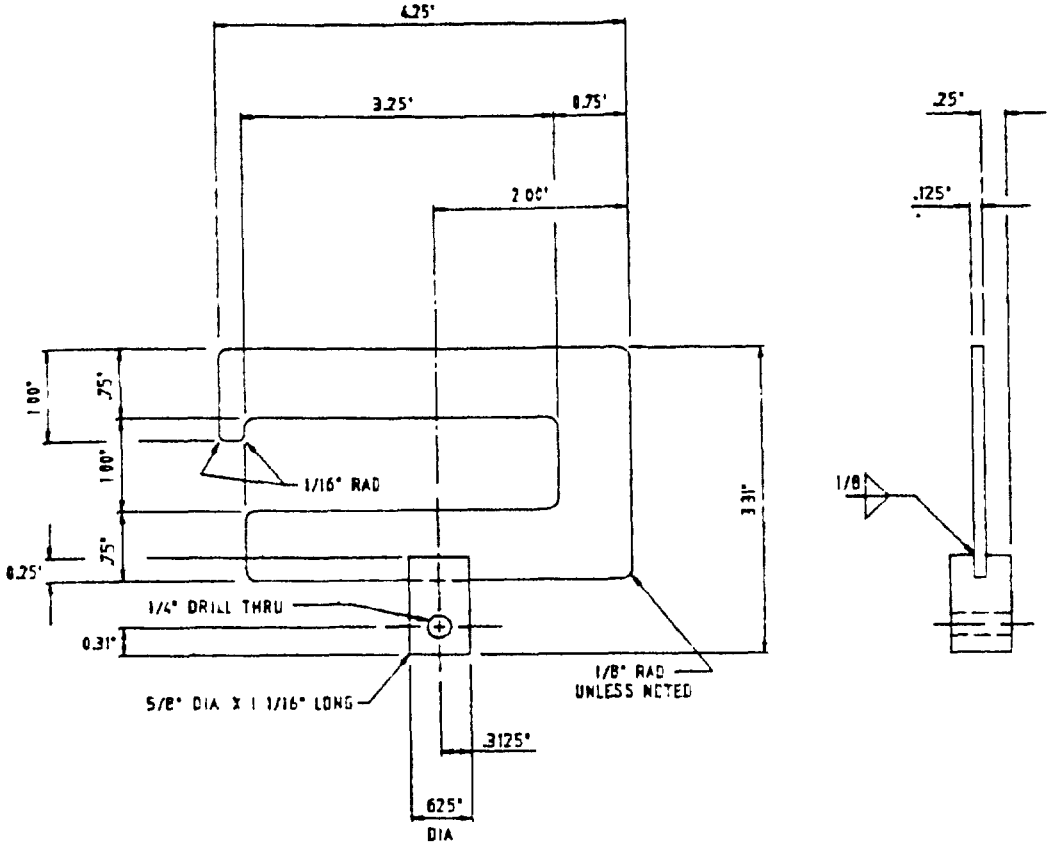



FIGURE 3 C-Clamp Dimensions - Small Shaft



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-29567 | Cloth, Laminated, Stretchable |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION (Mark one) |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER (Specify) |
| b. ADDRESS (Street, City, State, ZIP Code) | |

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER (Last, First, MI) – Optional | b. WORK TELEPHONE NUMBER (Include Area Code) – Optional |
|---|---|
| c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional | 8. DATE OF SUBMISSION (YYMMDD) |

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